

PRAYAS JEE 2.0 (2024)

Quadratic Equations

Revision Practice Sheet

JEE

Single Correct Type Questions (1 to 15)

1. The number of roots of the equation

$$\sqrt{x-2}(x^2-4x+3)=0 \text{ is}$$

- (1) Three (2) Four
(3) One (4) Two

2. The value of

$$3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$$

is equal to

- (1) $1.5 + \sqrt{3}$
(2) $2 + \sqrt{3}$
(3) $3 + 2\sqrt{3}$
(4) $4 + \sqrt{3}$

3. The sum of all the real values of x satisfying the equation $2^{(x-1)(x^2+5x-50)} = 1$ is:

- (1) 16
(2) 14
(3) -4
(4) -5

4. If both the roots of the quadratic equation $x^2 - 2px + p^2 + p - 5 = 0$ are less than 3, then k lies in the interval

- (1) $(-\infty, 1)$ (2) $(1, \infty)$
(3) $(1, 3)$ (4) $(3, \infty)$

5. For what values of the parameter k in the inequality

$$\left| \frac{x^2 + kx + 1}{x^2 + x + 1} \right| < 3, \text{ satisfied}$$

- (1) $-1 < k < 5$
(2) $1 < k < 5$
(3) Either
(4) None

6. If the quadratic equations $ax^2 + 2cx + b = 0$ and $ax^2 + 2bx + c = 0$ ($b \neq c$) have a common root, then $a + 4b + 4c$ is equal to

- (1) -2 (2) -1
(3) 0 (4) 1

7. If α, β are real and $\alpha^2, -\beta^2$ are the roots of $a^2x^2 + x + 1 - a^2 = 0$; ($a > 1$), then $\beta^2 =$

- (1) a^2 (2) 1
(3) $1 - a^2$ (4) $1 + a^2$

8. The equation $\sin^4 x - 2 \cos^2 x + a^2 = 0$ is solvable if

- (1) $-\sqrt{3} \leq a \leq \sqrt{3}$
(2) $-\sqrt{2} \leq a \leq \sqrt{2}$
(3) $-1 \leq a \leq 1$
(4) None of these

9. The value of a for which the quadratic equation $3x^2 + 2ax + 2x + a^2 - 3a + 2 = 0$ possesses roots of opposite signs lies in

- (1) $(-\infty, 1)$ (2) $(-\infty, 0)$
(3) $(1, 2)$ (4) $\left(\frac{3}{2}, 2\right)$

10. If α and β ($\alpha < \beta$), are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then

- (1) $0 < \alpha < \beta$ (2) $\alpha < 0 < \beta < |\alpha|$
(3) $\alpha < \beta < 0$ (4) $\alpha < 0 < |\alpha| < \beta$

11. If the roots of the equation $x^3 + 3px^2 + 3qx + r = 0$ are in A. P, then the condition is:

- (1) $2p^3 = 3pq + r$ (2) $2p^3 = 3pq$
(3) $2p^3 + r = 3pq$ (4) $2p^3 - r = -3pq$

12. If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots, then the value of $(a + b)$ is:

- (1) -4 (2) 2
(3) 6 (4) Cannot be determined



13. If α, β be the roots of $4x^2 - 17x + \lambda = 0$, $\lambda \in R$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral values of λ is

- (1) 1 (2) 2
(3) 3 (4) 4

14. The set of all real numbers x for which $x^2 - |x + 2| + x > 0$ is

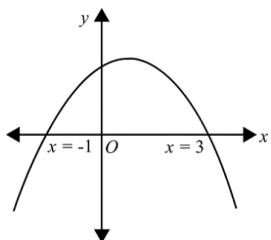
- (1) $(-\infty, -2) \cup (2, \infty)$
(2) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(3) $(-\infty, -1) \cup (1, \infty)$
(4) $(\sqrt{2}, \infty)$

15. If λ_1 and λ_2 are the two values of λ such that the roots α and β of the quadratic equation, $\lambda(x^2 - x) + x + 5 = 0$ satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \frac{4}{5} = 0$, then $\frac{\lambda_1}{\lambda_2^2} + \frac{\lambda_2}{\lambda_1^2}$ equal to:

- (1) 536 (2) 512
(3) 504 (4) 488

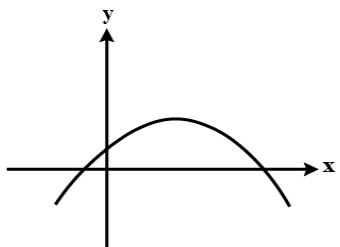
One or More Than One Type Questions (16 to 18)

16. Consider the graph of quadratic polynomial $y = ax^2 + bx + c$ as shown below. Which of the following is(are) correct?



- (1) $\frac{a-b+c}{abc} = 0$ (2) $abc(9a + 3b + c) < 0$
(3) $\frac{a+3b+9c}{abc} < 0$ (4) $ab(a - 3b + 9c) > 0$

17. The given figure shows the graph of $f(x) = ax^2 + bx + c$, then



- (1) $ac < 0$ (2) $bc > 0$
(3) $ab > 0$ (4) $abc < 0$

18. For any real x , the expression $2(k-x) \left[x + \sqrt{x^2 + k^2} \right]$

can exceed

- (1) k^2 (2) k
(3) $3k^2$ (4) None of these

Paragraph Type Questions (19 to 20)

Consider two quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$. Let α be the root that is common in the quadratic equation.

Then, $a_1\alpha^2 + b_1\alpha + c_1 = 0$ and $a_2\alpha^2 + b_2\alpha + c_2 = 0$

Using cramer's rule,

$$\frac{\alpha^2}{\begin{vmatrix} -c_1 & b_1 \\ -c_2 & b_2 \end{vmatrix}} = \frac{\alpha}{\begin{vmatrix} a_1 & -c_1 \\ a_2 & -c_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

The condition for only one common root is $(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$. If both roots are common, then the condition is $(a_1/a_2) = (b_1/b_2) = (c_1/c_2)$.

19. Find the value of p if the equation $3x^2 - 2x + p = 0$ and $6x^2 - 17x + 12 = 0$ have a common root.

- (1) $-8/3$ (2) $-15/3$
(3) 6 (4) $7/3$

20. If a, b, c belong to R and equations $ax^2 + bx + c = 0$ and $x^2 + 2x + 9 = 0$ have a common root, then $a : b : c = ?$

- (1) 1 : 2 : 3 (2) 2 : 1 : 9
(3) 1 : 2 : 9 (4) 9 : 2 : 1

Match the Column Type Ques. 21 to 22

21.

	Column - I		Column - II
I.	Ramesh and Mahesh solve an equation. In solving Ramesh commits a mistake in constant term and finds the root 8 and 2. Mahesh commits a mistake in the coefficient of x and finds the roots -9 and -1 . Find the sum of correct roots.	P.	7



II.	α & β are the roots of the equation $x^2 - 7x - 1 = 0$, then $\frac{\alpha^{10} + \beta^{10} - (\alpha^8 - \beta^8)}{\alpha^{\alpha+\beta+2} - \beta^{\alpha+\beta+2}}$	Q.	114
III.	The minimum value of $(x-6)^2 + (x+3)^2 + (x-8)^2 + (x+4)^2 + (x-3)^2$ is	R.	10
IV.	The number of integral values of k for which the equation $3 \sin x + 4 \cos x = k + 1$ has a solution, $k \in R$ is	S.	11

- (1) I-P, II-S, III-R, IV-Q
 (2) I-S, II-R, III-P, IV-Q
 (3) I-R, II-P, III-Q, IV-S
 (4) I-R, II-P, III-S, IV-Q

22.

	Column – I		Column – II
I.	Let $\lambda \neq 0$ be the real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation with sum of roots as.	P.	0

II.	The number of real roots of the equation $\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0,$ is	Q.	1
III.	The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has solutions in R. (R stands for real numbers)	R.	5

- (1) I-P, II-R, III-Q
 (2) I-Q, II-R, III-P
 (3) I-P, II-Q, III-R
 (4) I-R, II-P, III-Q

Integer Type Que. Nos. 23 to 25

23. Number of real solutions of the equation $5 + |2^x - 1| = 2^x(2^x - 2)$ is?
24. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is:
25. Let α, β ($\alpha > \beta$) be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n$, $zn \in N$, then $(P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15})/P_{13}P_{14}$ is equal to _____.



Hints and Solutions

1. (4)

2. (1)

3. (3)

4. (1)

5. (1)

6. (3)

7. (2)

8. (2)

9. (3)

10. (2)

11. (3)

12. (2)

13. (2)

14. (2)

15. (4)

16. (1,3)

17. (1, 2,4)

18. (1,2)

19. (1)

Let a be the common root of the equations.
Then a will satisfy both the equation.

Thus,

$$3a^2 - 2a + p = 0$$

$$6a^2 - 17a + 12 = 0$$

By cross multiplication we have,

$$a^2/(-24 + 17p) = a/(6p - 36) = -1/39$$

From first and second we have

$$a = (17p - 24)/(6p - 36) \dots\dots\dots(1)$$

From second and third, we have

$$a = (6p - 36)/(-39)$$

$$\text{or, } a = (2p - 12)/(-13) \dots\dots\dots(2)$$

From (1) and (2) we get

$$(17p - 24)/(6p - 36) = (2p - 12)/(-13)$$

$$\text{or, } 12p^2 - 144p + 432 = -221p + 312$$

$$\text{or, } 12p^2 + 77p + 120 = 0$$

$$\dots\dots\dots(3)$$

$$\text{or, } 12p^2 + 45p + 32p + 120 = 0$$

$$\text{or, } 3p(4p + 15) + 8(4p + 15) = 0$$

$$\text{or, } (3p + 8)(4p + 15) = 0$$

$$\text{or, } p = -8/3, -15/4$$

20. (3)

The roots of the equation $x^2 + 2x + 9 = 0$ are imaginary as the discriminant $D = 4 - 36 = -32$, which is negative.

As the imaginary roots of a quadratic are conjugate of each other, so both the roots of the equation will be common.

$$\text{So, } a/1 = b/2 = c/9$$

$$\text{or, } a:b:c = 1:2:9$$

21. (3)

22. (4)

23. (1)

Case I

$$2^x \geq 1$$

$$\Rightarrow 5 + 2^x - 1 = 2^x(2^x - 2)$$

$$\text{Let } 2^x = t$$

$$\Rightarrow 5 + t - 1 = t(t - 2)$$

$$= t^2 - 3t - 4 = 0$$

$$\Rightarrow t = 4 \text{ or } -1 \text{ (invalid) } (\because 2^x \neq -1)$$

$$\Rightarrow 2^x = 4$$

$$\Rightarrow x = 2$$

$\therefore$ only one solution

Case II

$$2^x < 1$$

$$\Rightarrow 5 + 1 - 2^x = 2^x(2^x - 2)$$

$$\Rightarrow t^2 - t - 6 = 0$$

$$\Rightarrow (t - 3)(t - 2) = 0$$

$$\Rightarrow t = 3, -2$$

$$2^x = 2 \text{ (invalid as } 2^x < 1), \text{ or } 2^x = -2 \text{ (invalid)}$$

Therefore, number of real roots is one.



24. (4)

Given:

$$x^2 - 3|x| + 2 = 0$$

$$\Rightarrow |x|^2 - 3|x| + 2 = 0 \quad (\because x^2 = |x|^2)$$

$$\Rightarrow |x|^2 - 2|x| - |x| + 2 = 0$$

$$\Rightarrow |x|(|x| - 2) - 1(|x| - 2) = 0$$

$$\Rightarrow (|x| - 2)(|x| - 1) = 0$$

$$\Rightarrow (|x| - 2) = 0 \text{ or } (|x| - 1) = 0$$

$$\Rightarrow |x| = 2 \text{ or } 1$$

$$\therefore x = \pm 2 \text{ or } \pm 1$$

25. (16)

$$P_n = \alpha^n - \beta^n \quad x^2 - x - 4 = 0$$

$$\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}} \dots\dots(1)$$

$$\text{As } P_n - P_{n-1} = (\alpha^n - \beta^n) - (\alpha^{n-1} - \beta^{n-1})$$

$$= \alpha^{n-2}(\alpha^2 - \alpha) - \beta^{n-2}(\beta^2 - \beta)$$

$$= 4(\alpha^{n-2} - \beta^{n-2})$$

$$P_n - P_{n-1} = 4 P_{n-2}$$

Hence Expression (1)

$$\frac{P_{16}(P_{15} - P_{14}) - P_{15}(P_{15} - P_{14})}{P_{13}P_{14}}$$

$$= \frac{(P_{15} - P_{14})(P_{16} - P_{15})}{P_{13}P_{14}}$$

$$= \frac{(4P_{13})(4P_{14})}{P_{13}P_{14}} = 16$$



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>